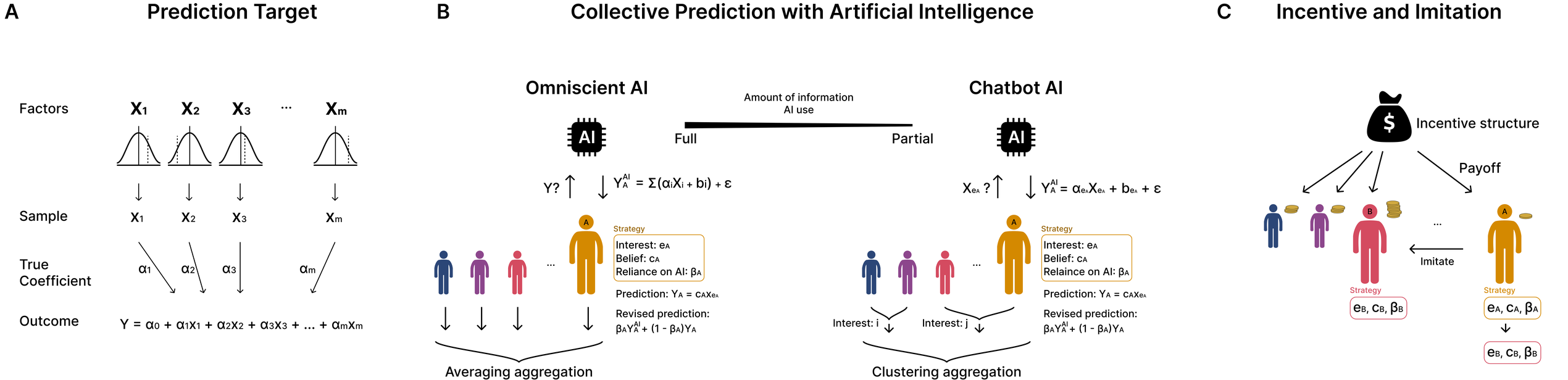
A
Prediction Target
Factors
X_1 X_2 X_3 ... X_m
Sample
x_1 x_2 x_3 x_m
True Coefficient
α_1 α_2 α_3 α_m
Outcome
$Y = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \ldots + \alpha_m x_m$
B
Collective Prediction with Artificial Intelligence
Omniscient AI
Amount of information AI use
Full
Partial
Chatbot AI
AI
$Y?$
$Y_A^{AI} = \Sigma(\alpha_i X_i + b_i) + \varepsilon$
$X_{e_A}?$
$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \varepsilon$
Strategy
Interest: e_A
Belief: c_A
Reliance on AI: β_A
Relaince on AI: β_A
Prediction: $Y_A = c_A x_{e_A}$
Revised prediction: $\beta_A Y_A^{AI} + (1 - \beta_A) Y_A$
Interest: i
Interest: j
Averaging aggregation
Clustering aggregation
C
Incentive and Imitation
Incentive structure
Payoff
Imitate
Strategy
e_B, c_B, β_B
e_A, c_A, β_A